

AI/ML Task 2 — Model Comparison Charts

MainCrafts Technology — AI & ML Internship · Dataset: California Housing
(sklearn.datasets.fetch_california_housing) · Target: HousePrice

About this PDF

This document contains the three Excel charts built from Task2_Predictions.xlsx. Each chart is a native Excel chart bound to live cell ranges in that workbook (Predictions and Comparison sheets); LibreOffice rendered them to the pages that follow. The metrics table below is read from outputs/metrics.json — the same file the notebook writes at run time — so no number in this PDF was typed by hand.

Verified test-set metrics

Model	Test RMSE	Test R ²	Rank
Linear Regression	0.7456	0.5758	3
Ridge Regression	0.7456	0.5758	2
Decision Tree [best]	0.7242	0.5997	1

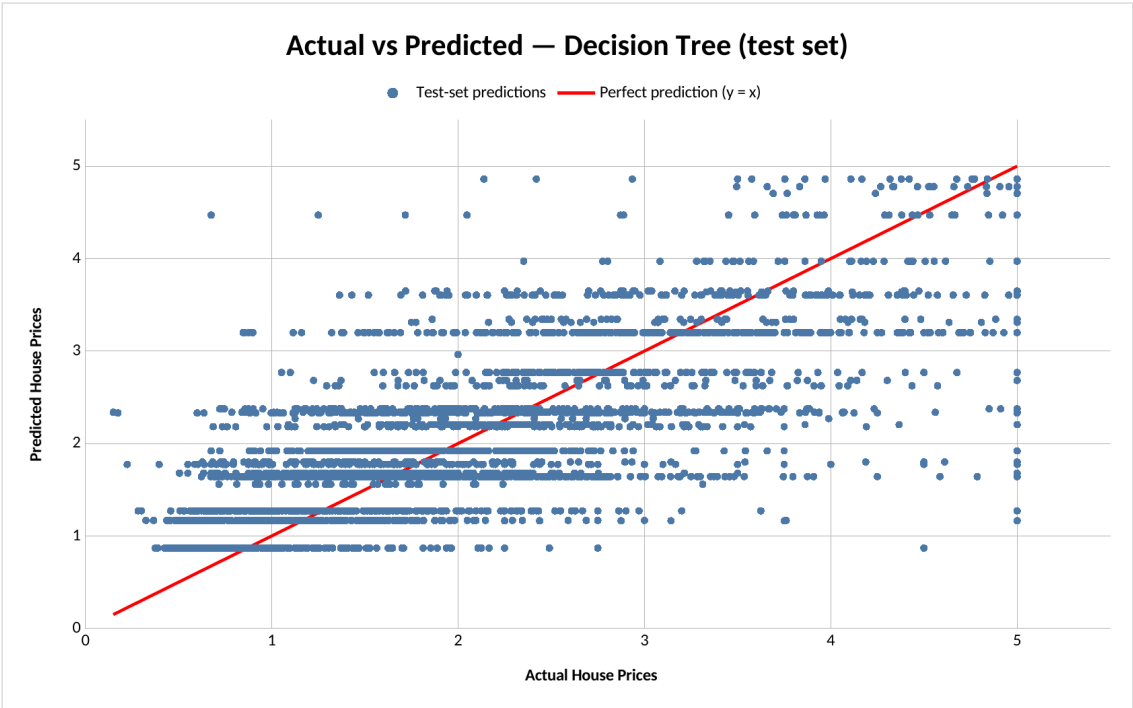
Run configuration

Dataset	California Housing via sklearn.datasets.fetch_california_housing(as_frame=True)
Target column	HousePrice
Rows train / test	16,512 / 4,128
test_size	0.2
random_state	42
Preprocessing	StandardScaler fit on X_train only, then transform train and test
Models	LinearRegression() · Ridge(alpha=1.0) · DecisionTreeRegressor(max_depth=5, random_state=42)
Selection rule	argmin(test RMSE); tiebreak argmax(test R ²) — computed programmatically
Chart source	Native Excel charts in Task2_Predictions.xlsx, rendered by LibreOffice

Pages 2–4 that follow show, in order: Chart 1 (Actual vs Predicted for the Decision Tree), Chart 2 (Residuals vs Predicted for the Decision Tree), Chart 3 (RMSE and R² across all three models).

Chart 1 — Actual vs Predicted (Decision Tree)

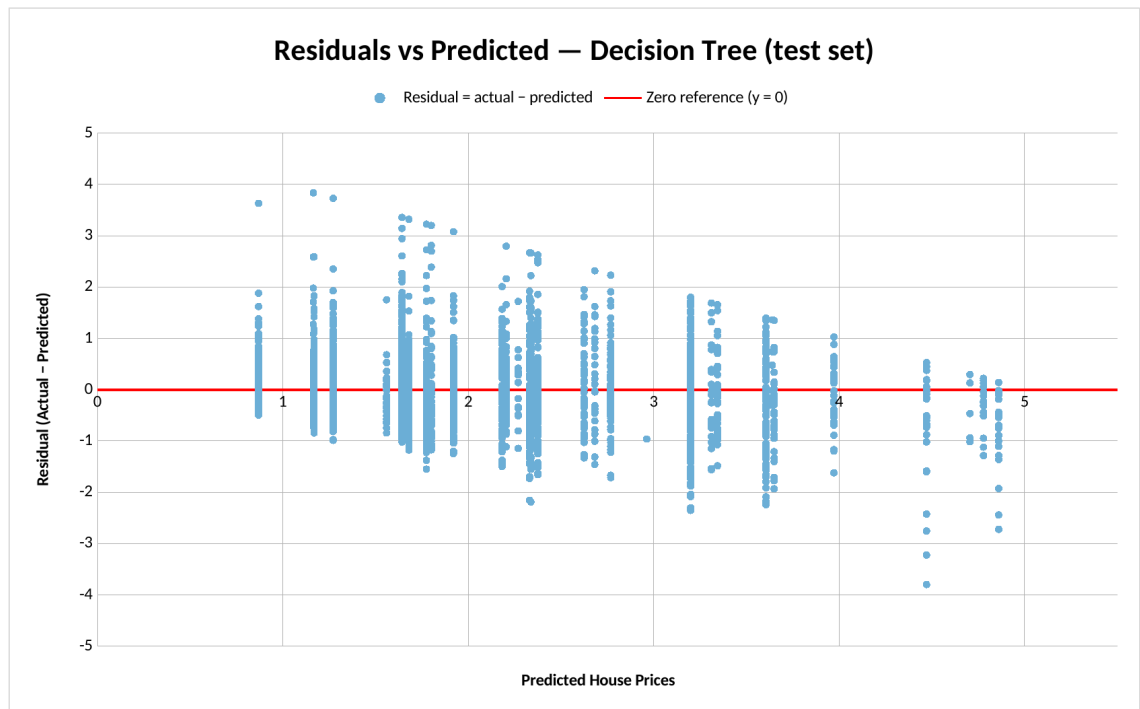
Chart 1 — Actual vs Predicted (Decision Tree). Data source: Predictions!B:B (Actual) vs Predictions!E:E (Decision_Tree_Predicted); reference line: YX_ReferenceLine!B2:C3.



Blue dots are the 4,128 test-set predictions from the winning model. The red line is $y = x$ (a perfect prediction). Horizontal banding appears because a depth-5 tree produces only 32 distinct leaf values.

Chart 2 — Residuals vs Predicted (Decision Tree)

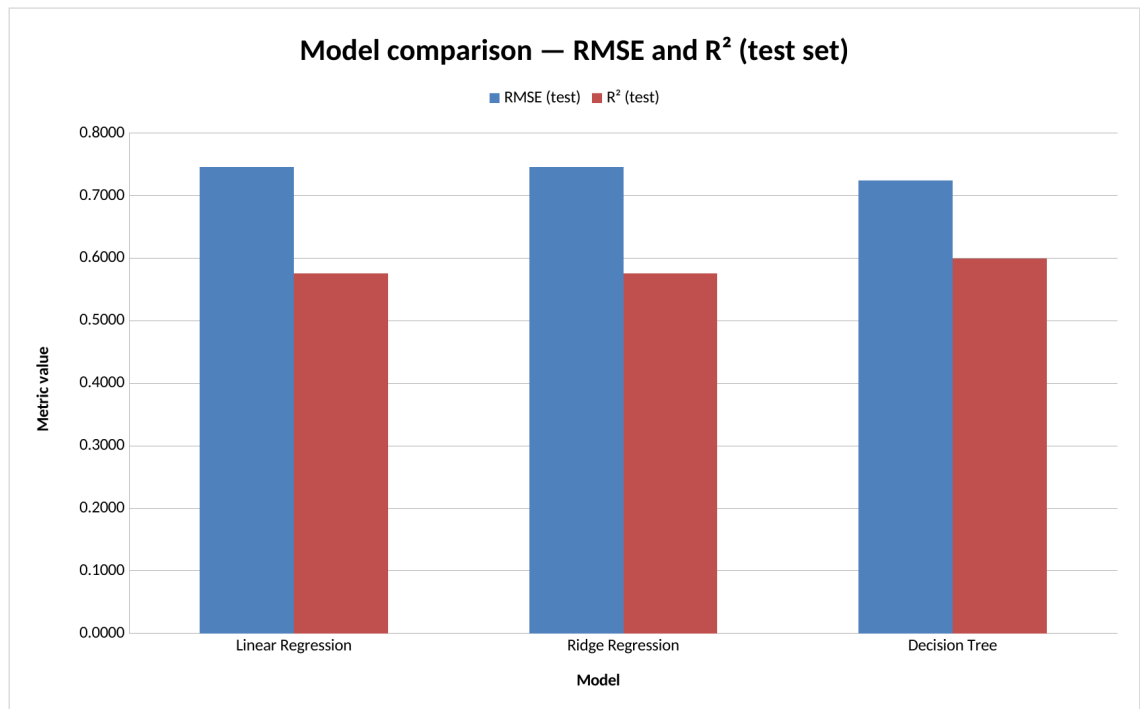
Chart 2 — Residuals vs Predicted (Decision Tree). Data source: Predictions!E:E (Decision_Tree_Predicted) vs Predictions!F:F (Residual_DT); zero line: YX_ReferenceLine!E2:F3.



Residual = Actual – Predicted. The red horizontal line is $y = 0$ (no error). Residuals cluster in vertical columns at the 32 leaf-mean predictions. Mean residual = -0.0019, standard deviation = 0.724 on the test set.

Chart 3 — Model comparison (RMSE and R² on test set)

Chart 3 — RMSE and R² by model. Data source: Comparison!A1:C4 (RMSE and R² formulas).



Side-by-side test RMSE (blue) and test R² (red) for the three models. Decision Tree has the lowest RMSE and highest R² of the three; the two linear models are practically indistinguishable at this precision.